



भारतीय सूचना प्रौद्योगिकी संस्थान नागपुर

Indian Institute of Information Technology Nagpur

An Institution of National Importance By An Act of Parliament

Title:

Hybrid Framework for Craquelure Detection and Virtual Restoration in Digitized Paintings

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Shreyas Khare(BT23ECI058)

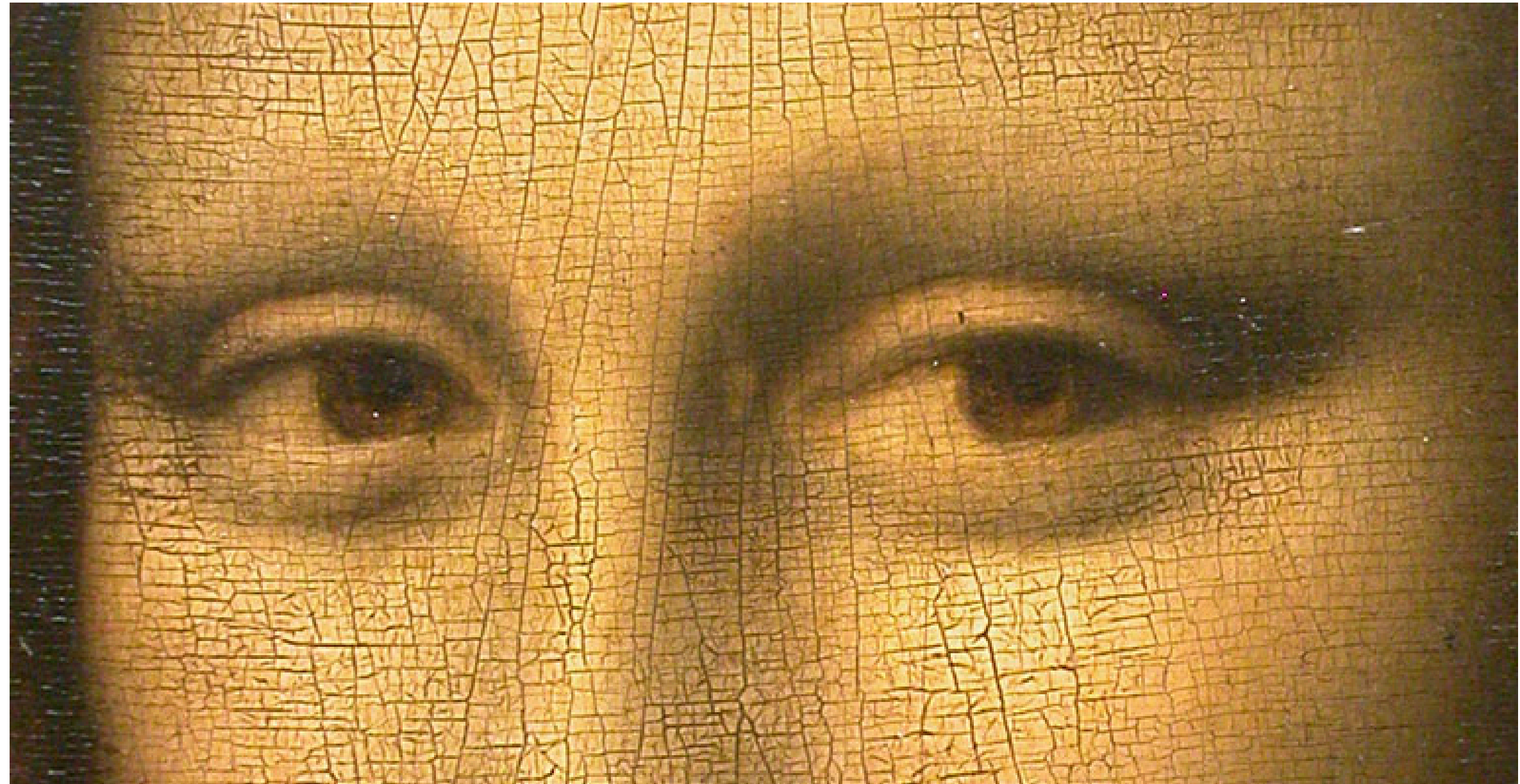
Presented to:- Dr. Tapan Kumar Jain



Why Restoration is Needed?

Old paintings develop cracks due to:

1. Aging
2. Drying of paint
3. Mechanical damage
 - Cracks reduce beauty and image quality
 - Manual restoration is expensive and risky
 - Digital restoration is safe and non-destructive

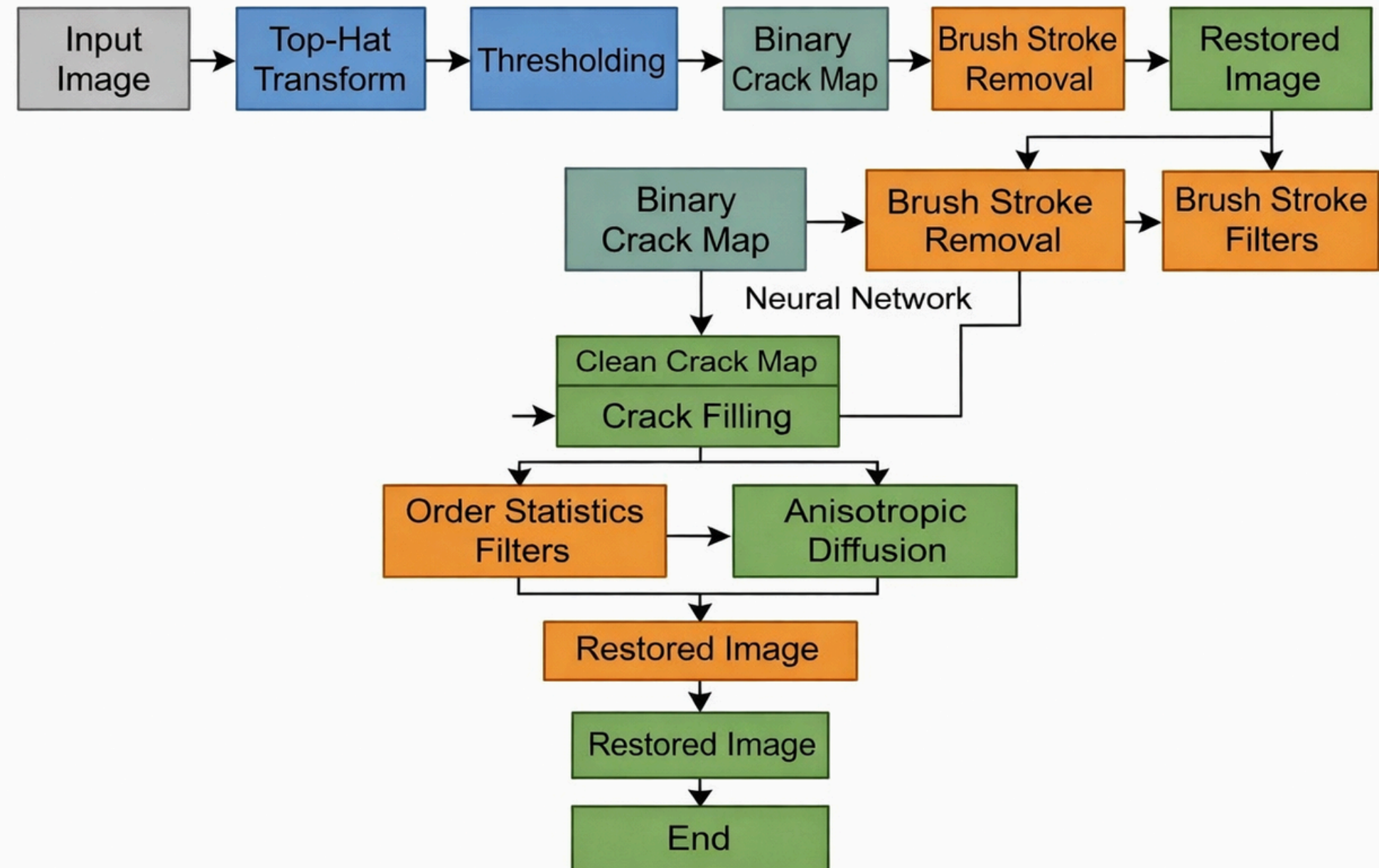


Objective

The paper proposes a complete system having 3 stages:

- Crack Detection
- Separation of brush strokes from cracks
- Crack Filling / Image Restoration

Crack Detection and Restoration Pipeline



Crack Detection Method

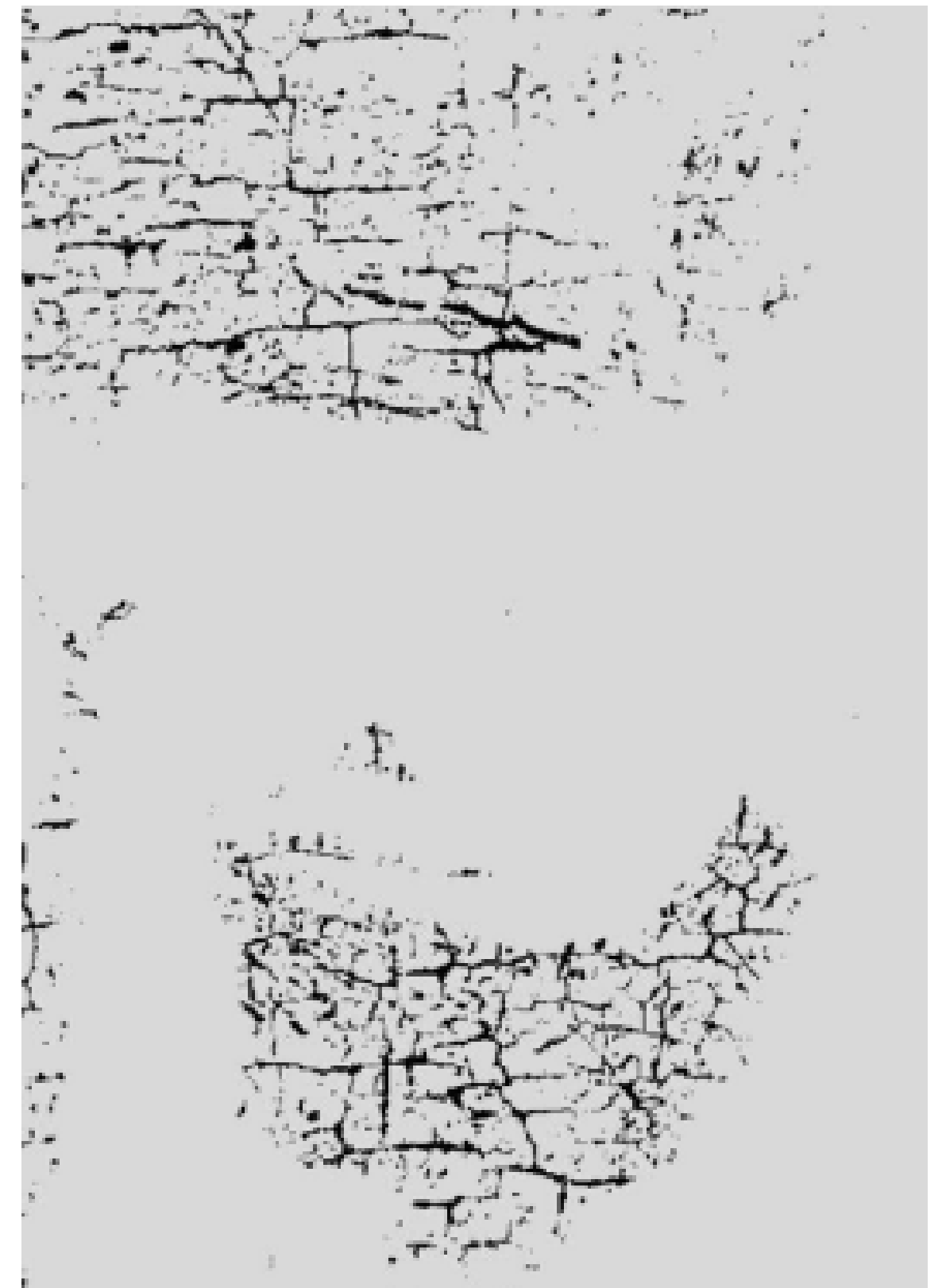
Stage 1: Crack Detection using Morphological Top-Hat Transform

- Cracks are dark thin lines
- They behave like local minima in image intensity
- We are using Top-Hat Transform
- Thresholding is applied after transform

$$T(f)=f-(f\circ b)$$

Where:

- f = image
- b = structuring element
- $\circ$ = opening of the original image



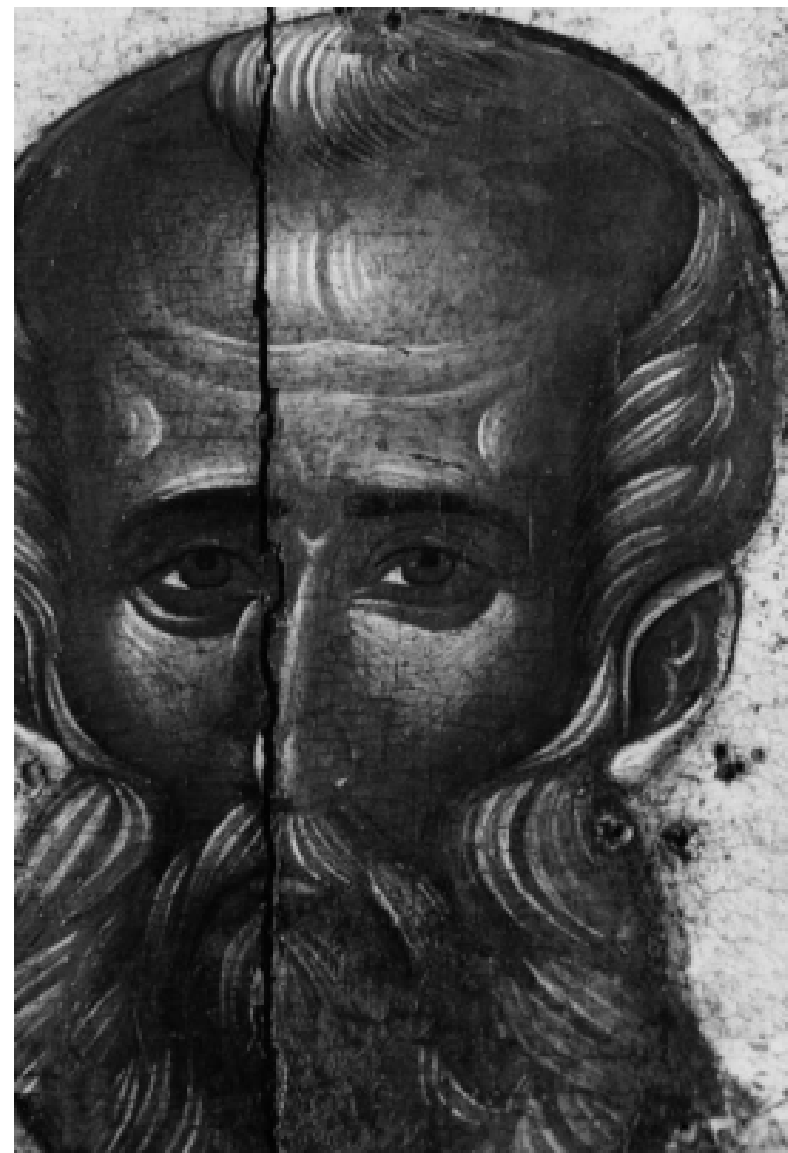
Problem: Brush Strokes Misdetected

Sometimes dark brush strokes look similar to cracks.

Examples:

- Hair
- Fine painting lines
- Borders

So system needs separation.



Separation Techniques

Two Methods Used:

1. Semi-Automatic Region Growing

- User selects true crack seeds
- Algorithm grows connected crack regions

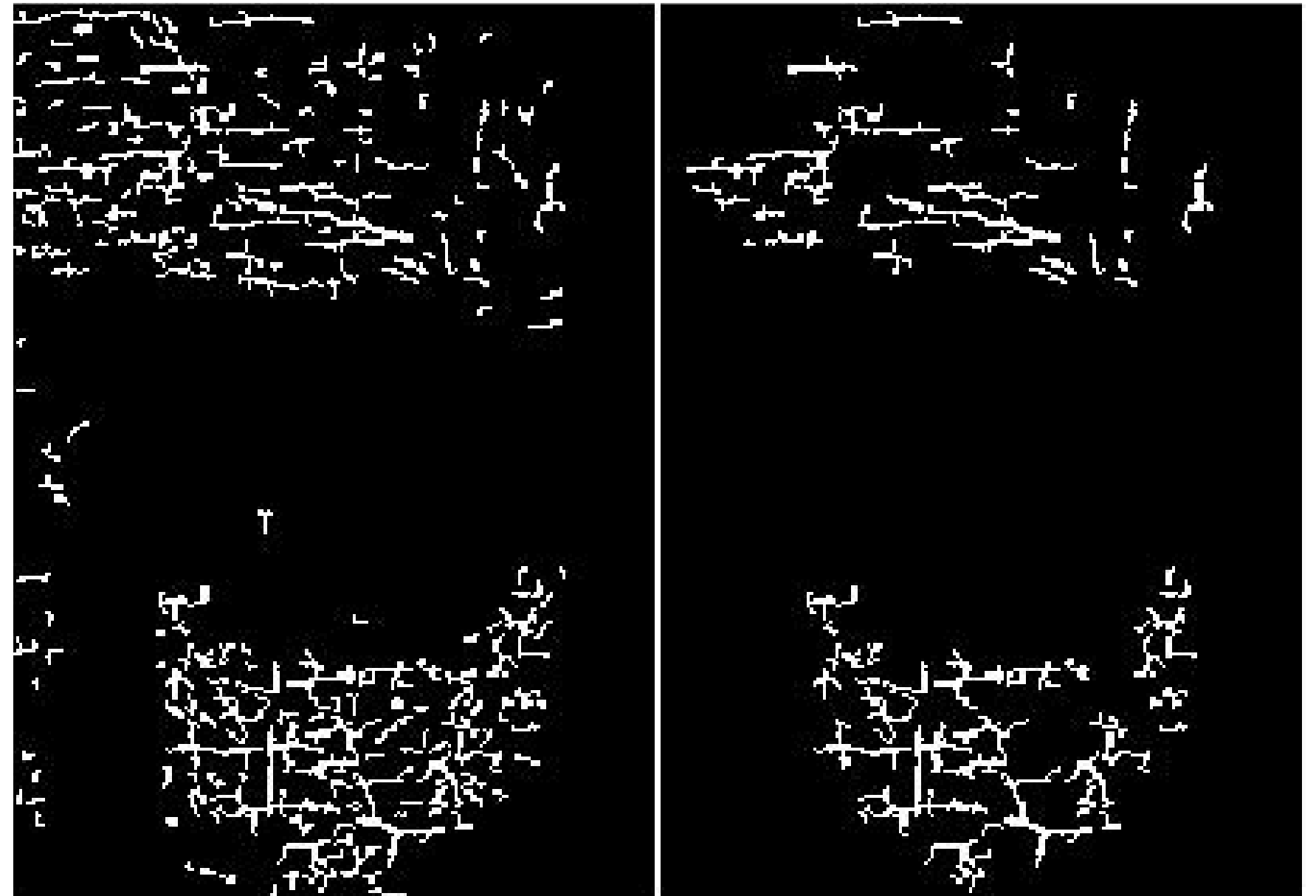
2. MRBF Neural Network

Uses:

- Hue
- Saturation

To classify:

- Crack pixels
- Brush stroke pixels



Crack Filling Methods

After detection, missing pixels are restored using:

1. Order Statistics Filters

- Median Filter
- Weighted Median
- Trimmed Mean Filter

2. Controlled Anisotropic Diffusion

- Orientation Sensitive Anisotropic Diffusion



Anisotropic Diffusion

This method:

- Smooths crack area
- Preserves edges/details
- Uses crack direction
- Fills only perpendicular to crack

$$\partial t / \partial I = \nabla \cdot (c(x, y, t) \nabla I)$$



Results

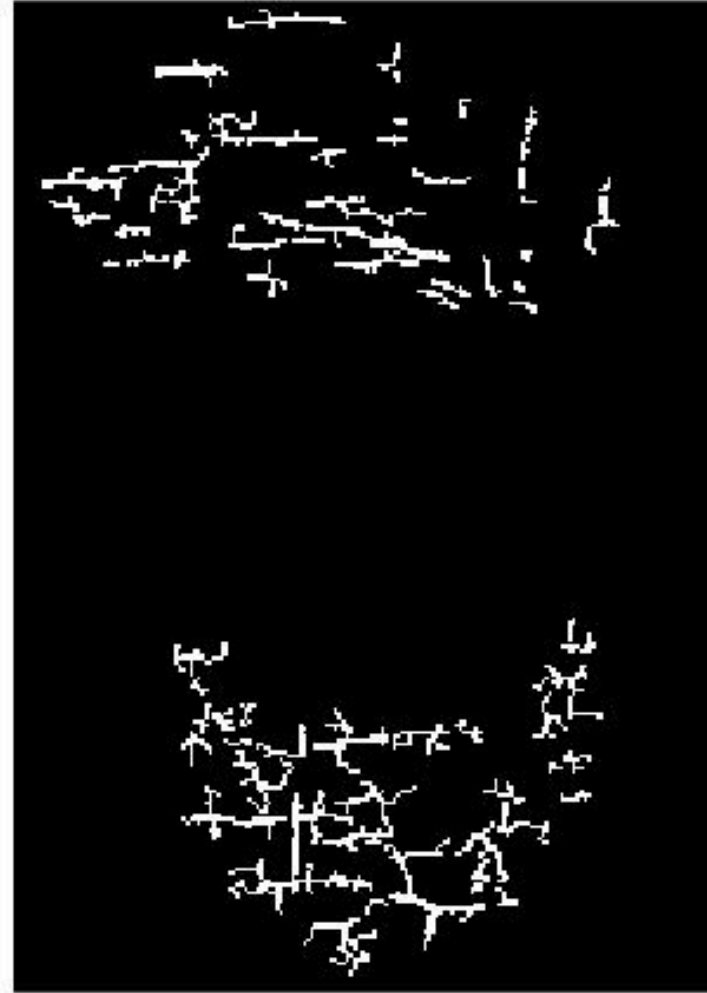
Step wise results are
as follows :

PART 2 — Comparison: Raw | Grassfire (2A) | MRBF (2B) | Combined

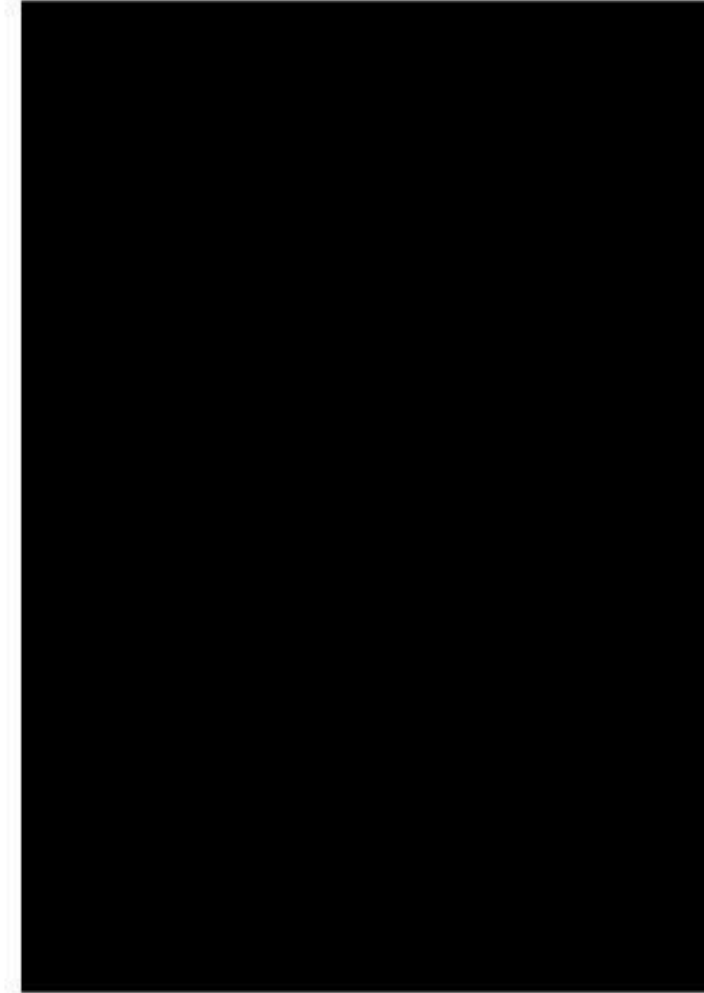
Raw Crack Map



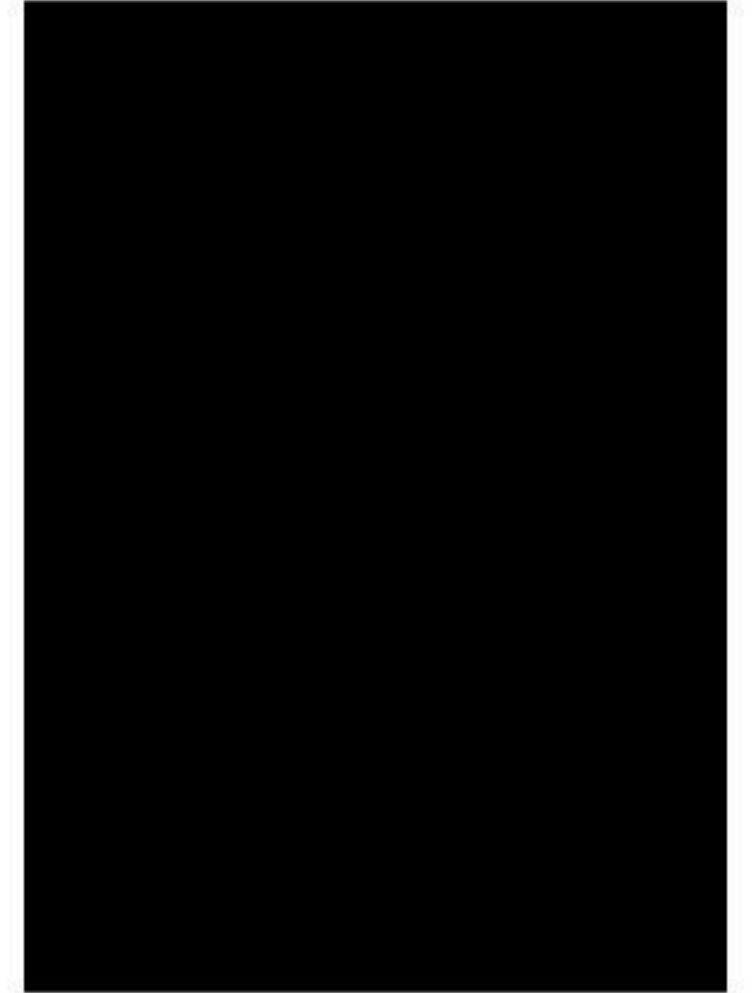
2A: Semi-Auto
(Grassfire)



2B: MRBF Neural Net
(Hue & Saturation)



Combined: 2A AND 2B
(Most Conservative)

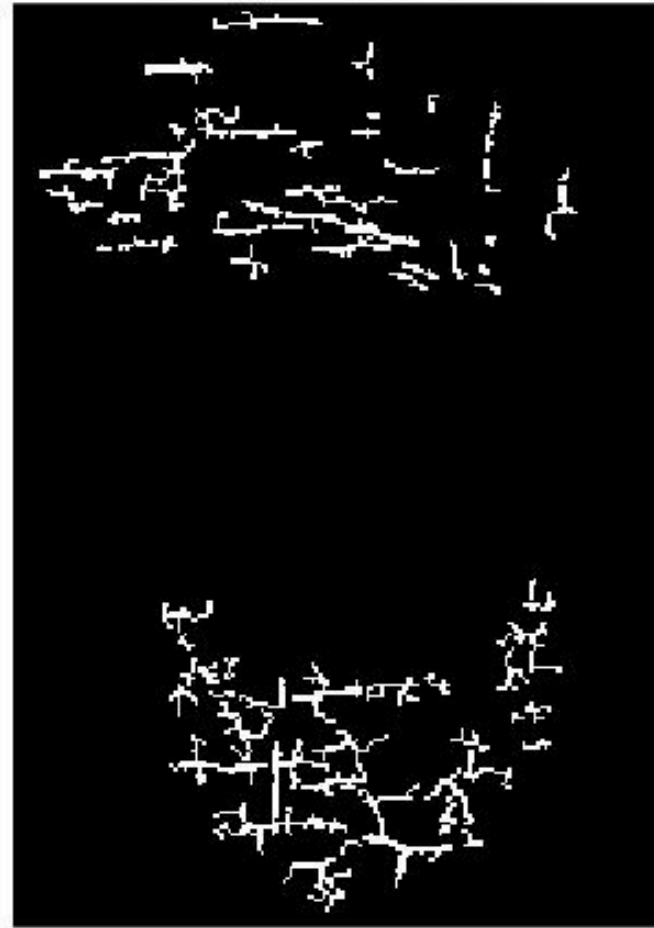


Final Comparison: Original vs Crack Mask vs MTM vs Anisotropic Diffusion

Original



Crack Mask (Part 2)



MTM Order Statistics
(Part 3A)



Anisotropic Diffusion
(Part 3B)



PART 1 — Crack Detection via Morphological Top-Hat Transform

1. Original Grayscale



2. Final Structuring Element



3. Dilation



4. Erosion



5. Opening



6. Closing



7. White Top-Hat



8. Black Top-Hat (BTH)



9. Binary Crack Map (T=18)



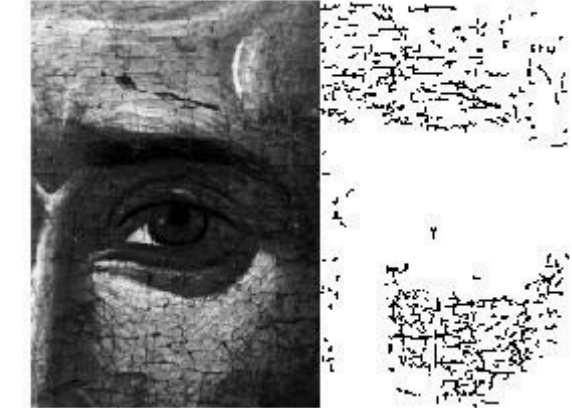
10. BW Crack Image



11. WTH vs BTH



12. Original vs Crack Image

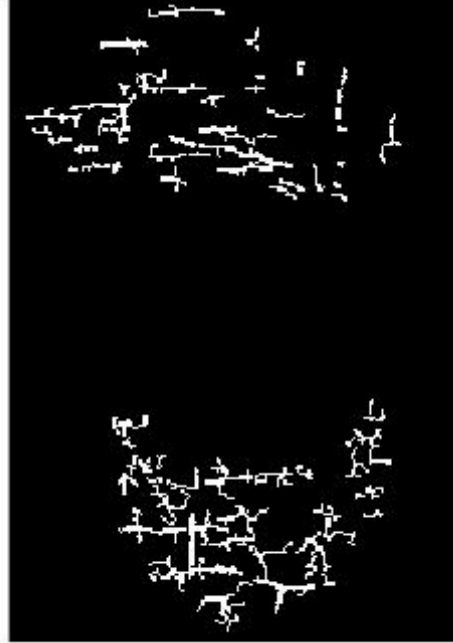


PART 3A — Crack Filling via Modified Trimmed Mean (Order Statistics) Filter

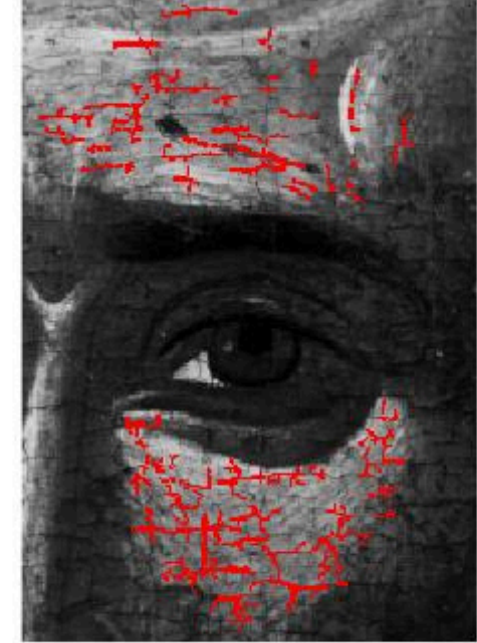
1. Original Grayscale



2. Final Crack Mask



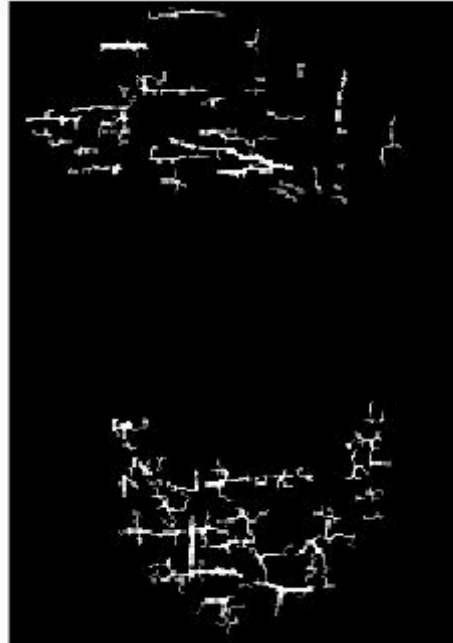
3. Cracks Highlighted (red)



4. Filled Image (MTM, win=5)



5. Difference Map ($\times 5$)



6. Original | MTM Filled

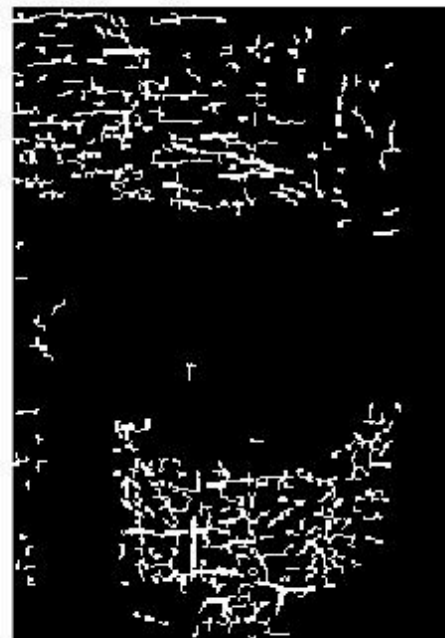


PART 2B — MRBF Neural Network Brush Stroke Removal (Hue & Saturation)

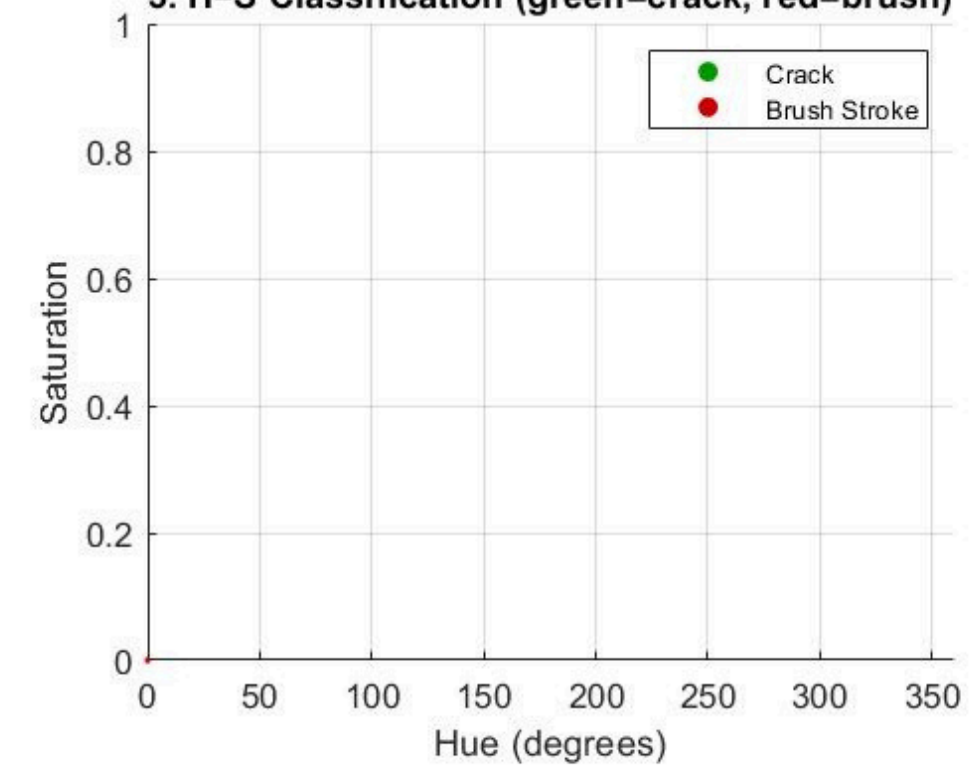
1. Original RGB Image



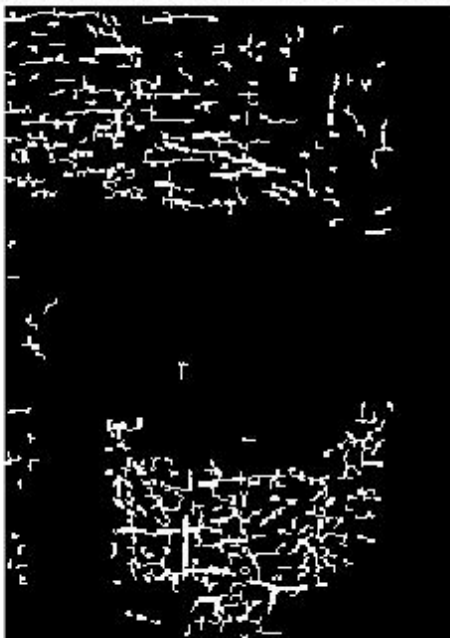
2. Raw Crack Map (from Part 1)



3. H-S Classification (green=crack, red=brush)



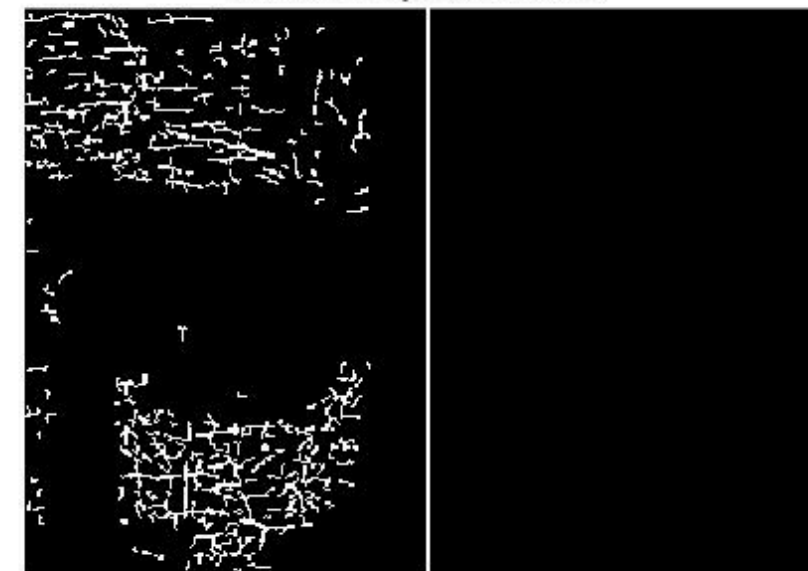
4. Removed Brush Strokes (MRBF)



5. Cleaned Crack Map (MRBF)



6. Before | After MRBF

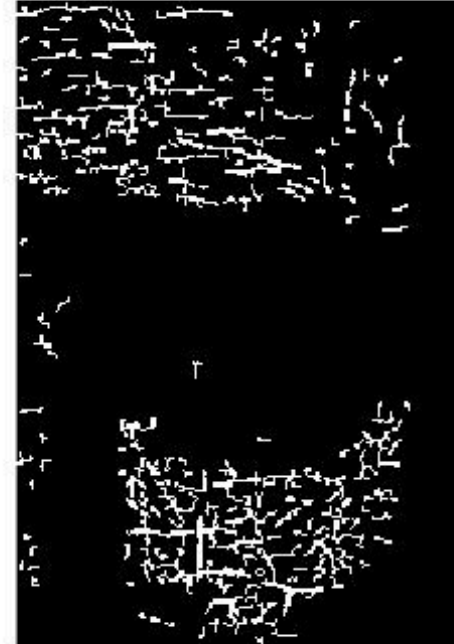


PART 2A — Semi-Automatic Brush Stroke Removal (Region-Growing / Grassfire)

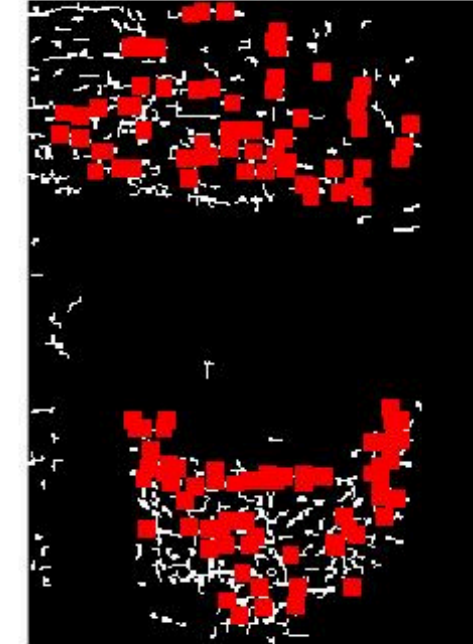
1. Original Grayscale



2. Raw Crack Map (from Part 1)



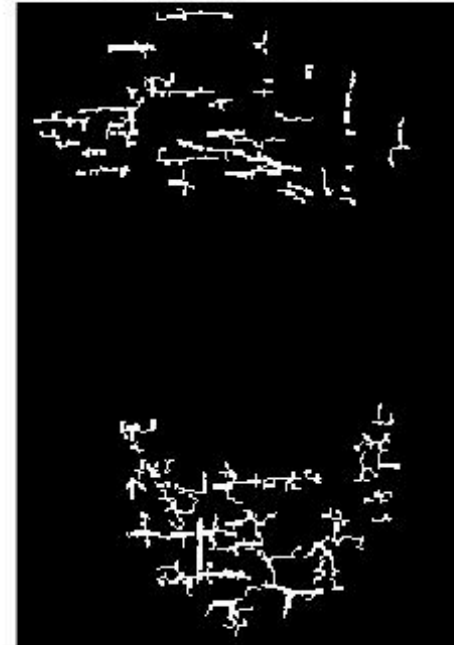
3. Crack Map with Seeds (red)



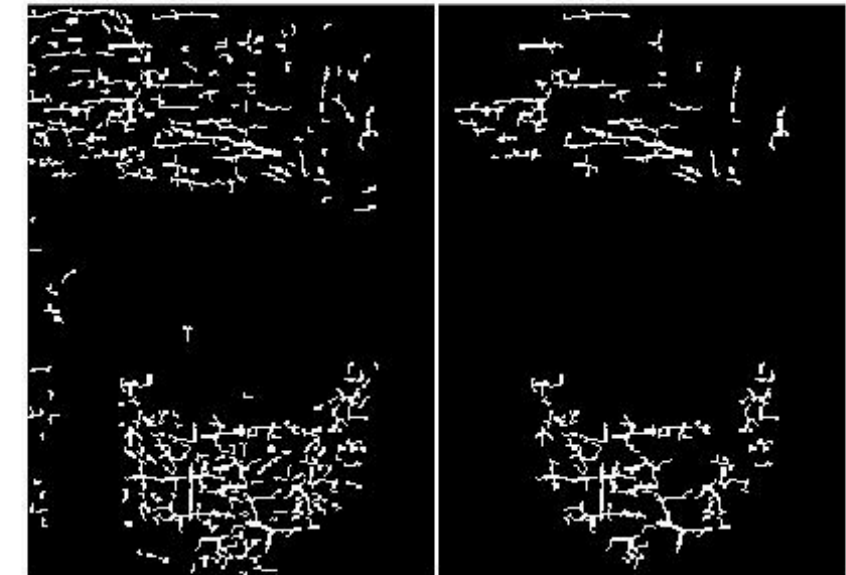
4. Removed Brush Strokes (2A)



5. Cleaned Crack Map (2A)



6. Before | After (2A)



Necessity of Part 2: Brush Stroke Removal Before Filling

Original



Part 3 on Raw Crack Map (no Part 2)



Part 3 after Part 2 (brush strokes removed)



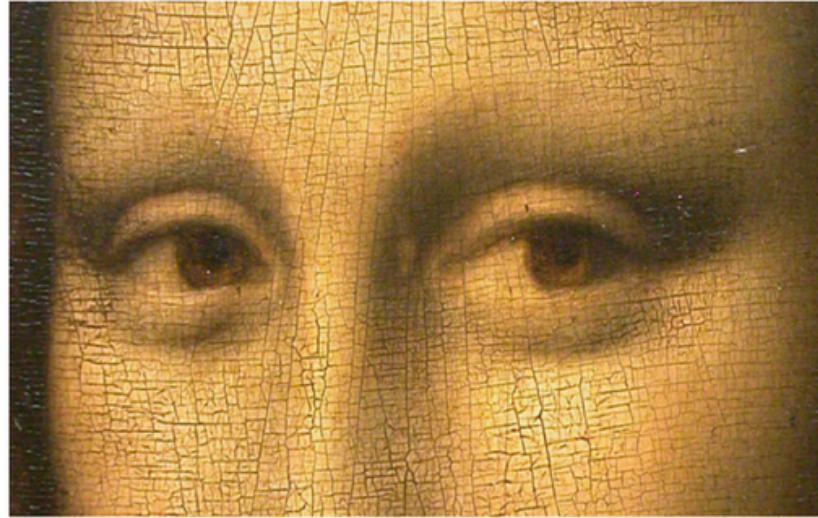
Modern Approach (Cuch-Guillén et al., 2026)

- Uses morphological Top-Hat + SegFormer transformer for crack detection
- Synthetic crack generator (Bézier curves) creates training data — no manual labeling needed
- Orientation-sensitive Anisotropic Diffusion for inpainting (50 iterations vs 20)
- Fully annotation-free pipeline

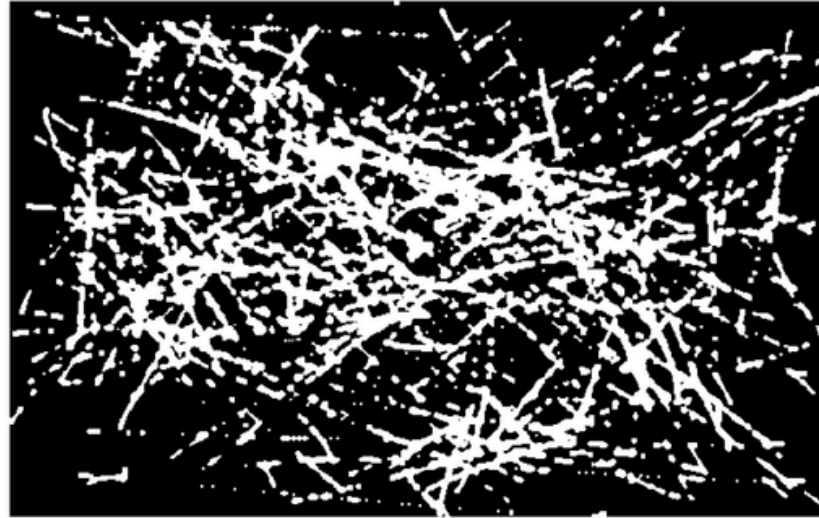


Synthetic Craquelure Generation — Bezier Curve Approach
(arXiv:2602.12742, Feb 2026)

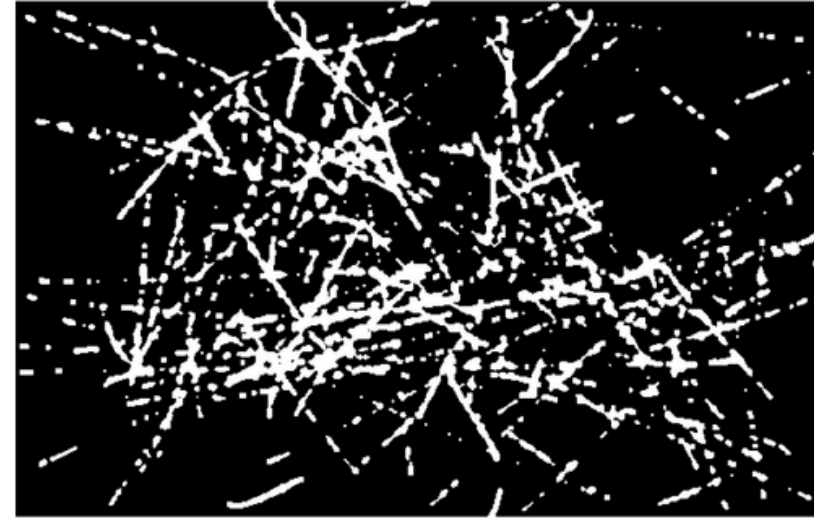
Clean Painting
(input)



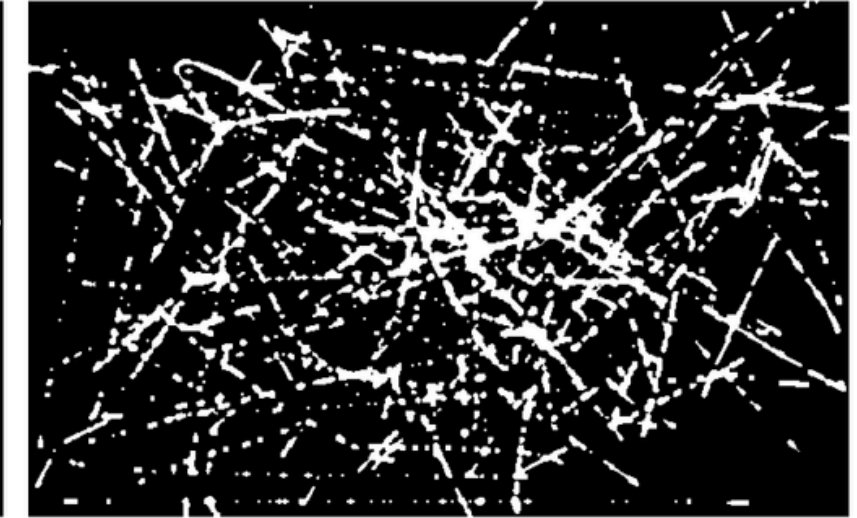
Synthetic Mask #1
(125 crack trajectories)



Synthetic Mask #2
(79 crack trajectories)



Synthetic Mask #3
(84 crack trajectories)



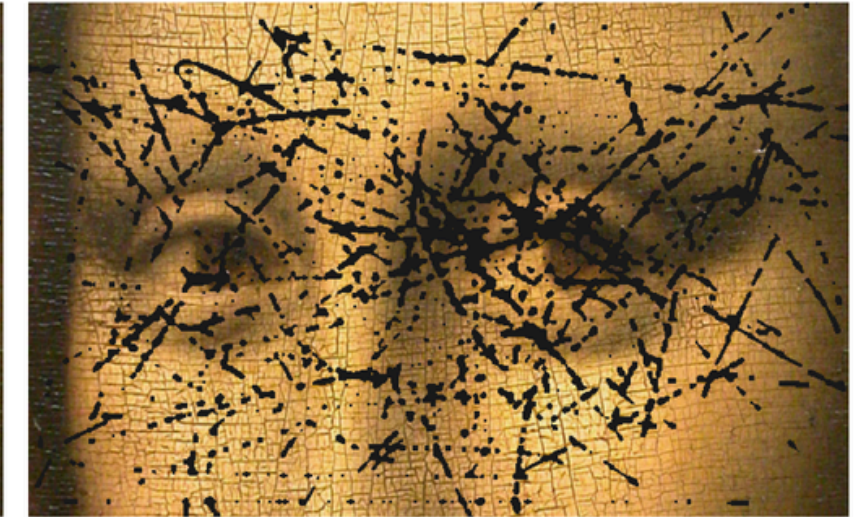
Damaged Image #1
(training sample)



Damaged Image #2
(training sample)



Damaged Image #3
(training sample)



Bezier Trajectories
+ Tapered Geometry
+ Branching
→ Realistic Cracks

Fig. Bézier crack generation

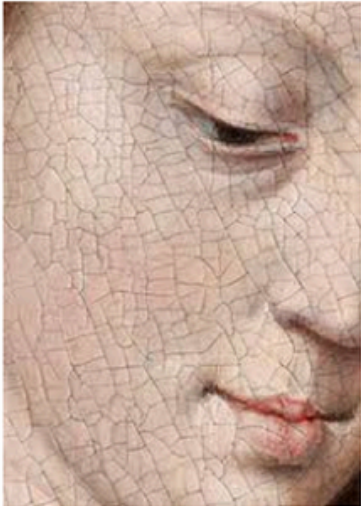
Modern Method Results

- Same top-hat detection as classical baseline
- Orientation-sensitive AD: 50 iterations, K=80 (vs classical K=127)
- Consistent improvement across all 4 test paintings
- Average SSIM improvement: +0.0025 over classical AD

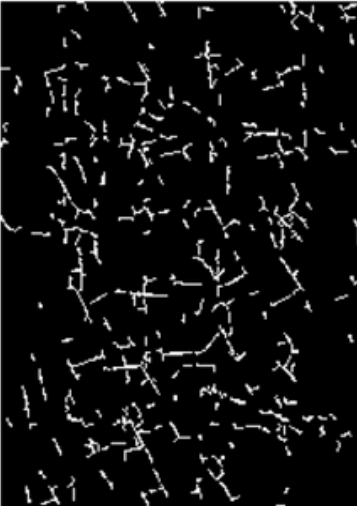
Method	PSNR (dB)	SSIM
Classical AD	30.87	0.9045
Modern AD	30.95	0.9061

Detection

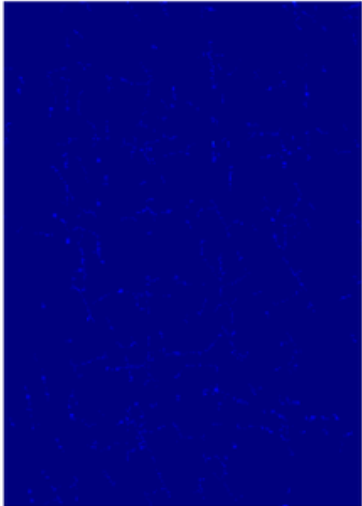
Original
(with cracks)



Shared Crack Mask
(Top-Hat Detection)



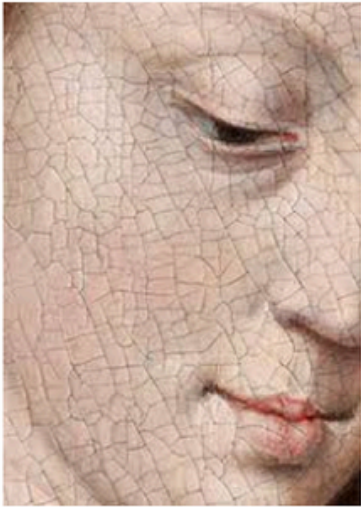
Difference Map
(Modern AD - Classical AD)



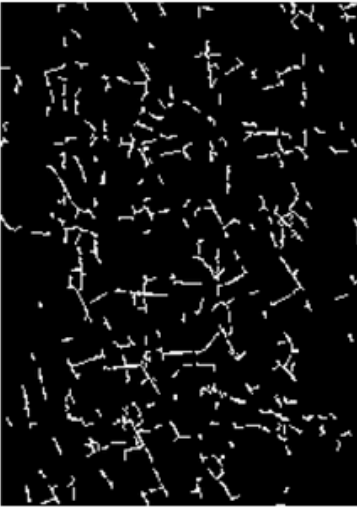
PSNR Comparison	
Classical MTM:	29.96 dB
Classical AD:	30.87 dB
Modern MTM:	30.43 dB
Modern AD:	30.95 dB
SSIM Comparison	
Classical AD:	0.9045
Modern AD:	0.9061
Improvement:	+0.0016

Classical
Restoration

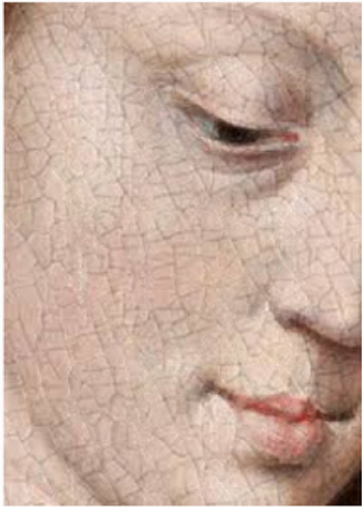
Original



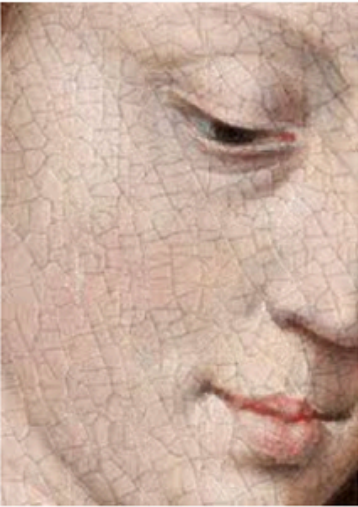
Crack Mask



Classical + MTM
PSNR=29.96 SSIM=0.883

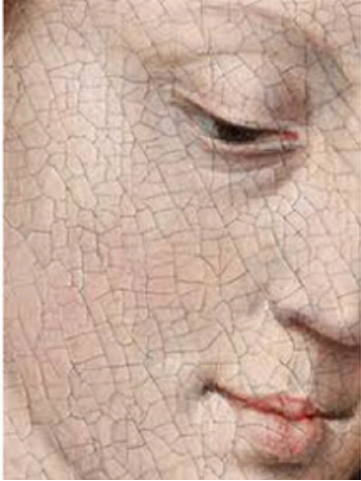


Classical + AD (20 iter)
PSNR=30.87 SSIM=0.904

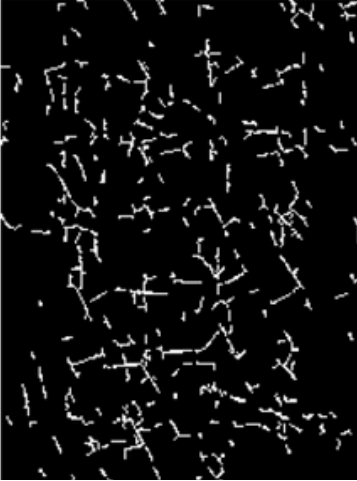


Modern
Restoration

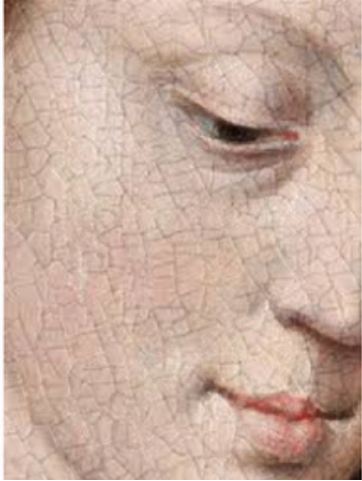
Original



Crack Mask



Modern + MTM (w=3)
PSNR=30.43 SSIM=0.894



Modern + Orient. AD (50 iter) (Best)
PSNR=30.95 SSIM=0.906

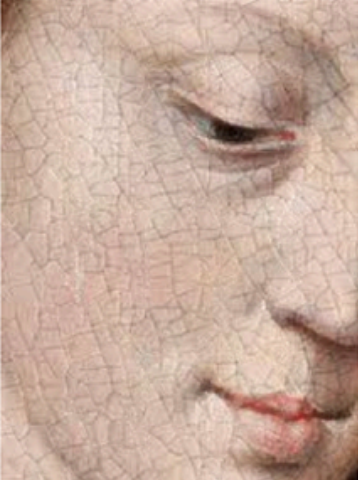


Fig. Comparison Result

Novel Method: Exemplar-Based Patch Inpainting (EBPI)

- Proposed method: finds best-matching texture patch from non-crack region
- Unlike AD (smooths locally), EBPI copies real texture from similar regions
- Limitation: PSNR measured vs cracked original — no ground truth available
- EBPI shows visually sharper fills but lower PSNR due to metric limitation

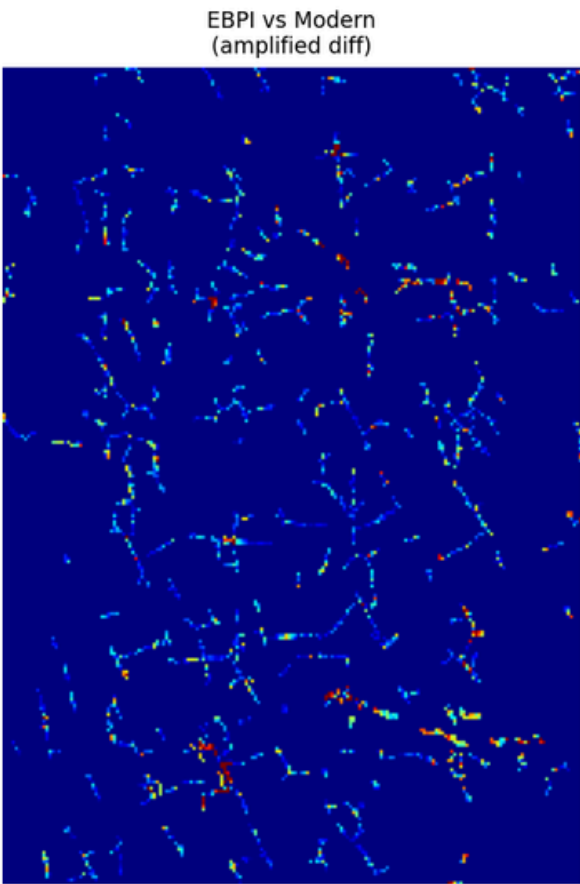
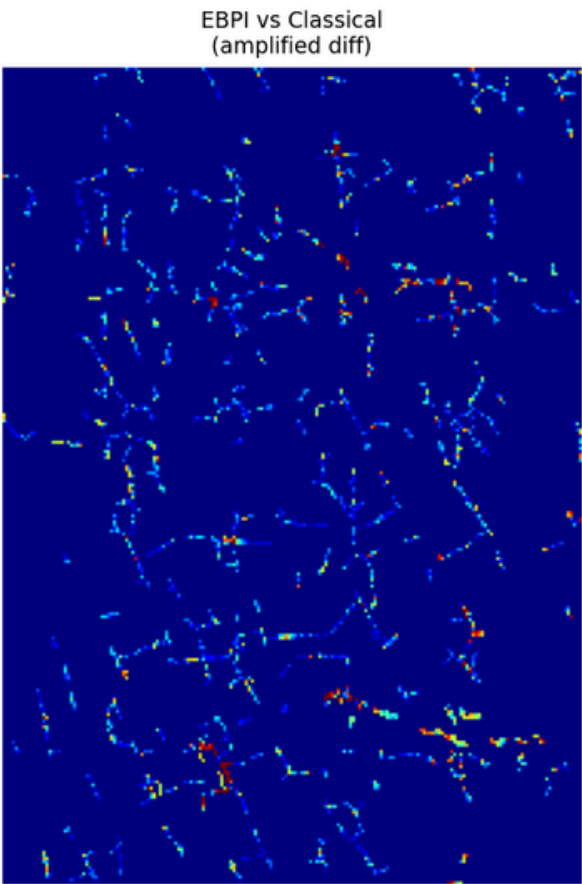
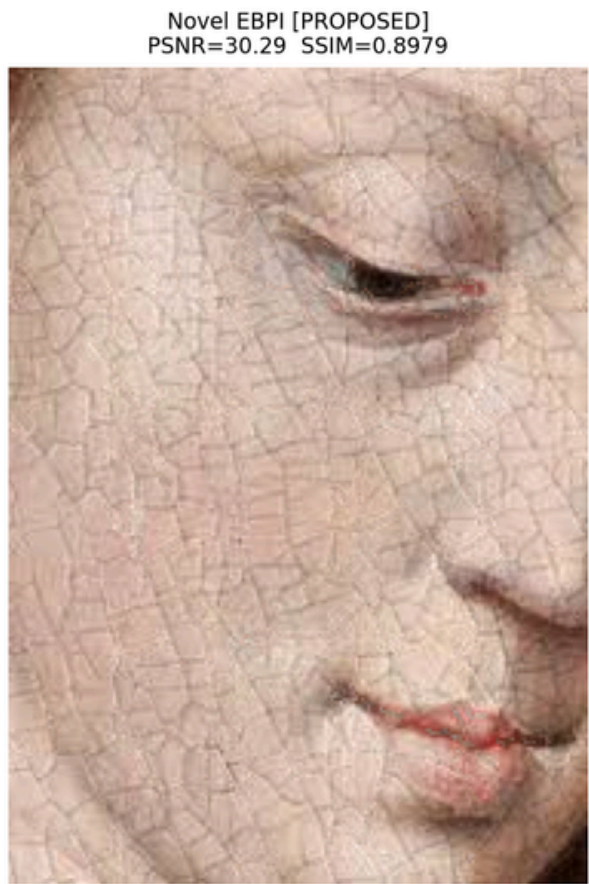
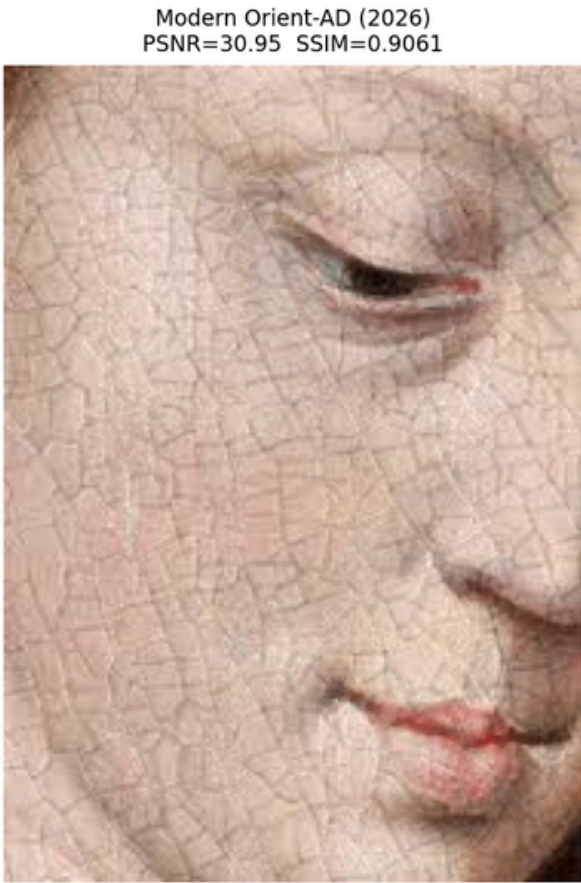
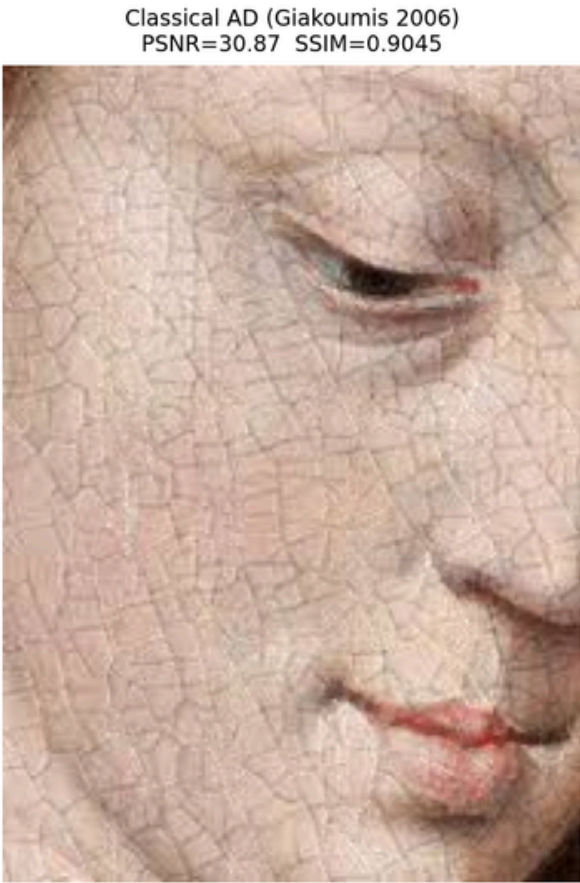
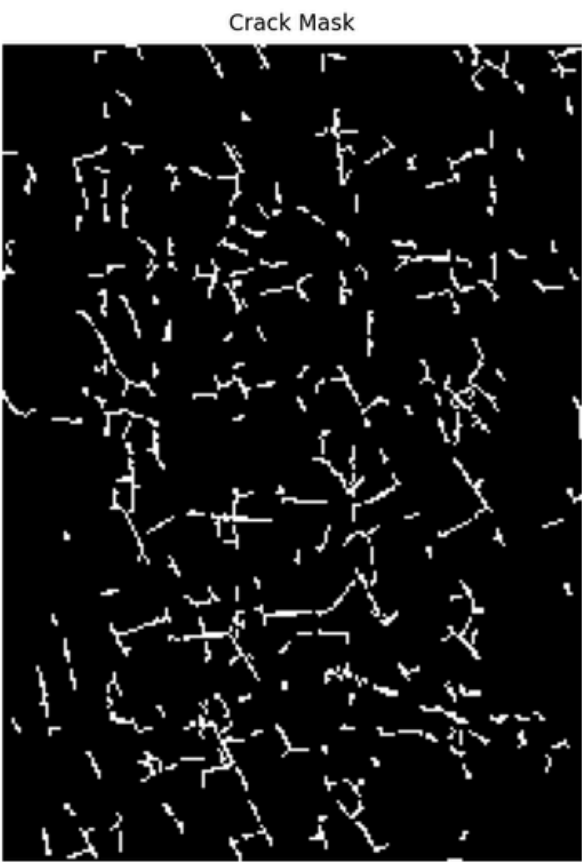
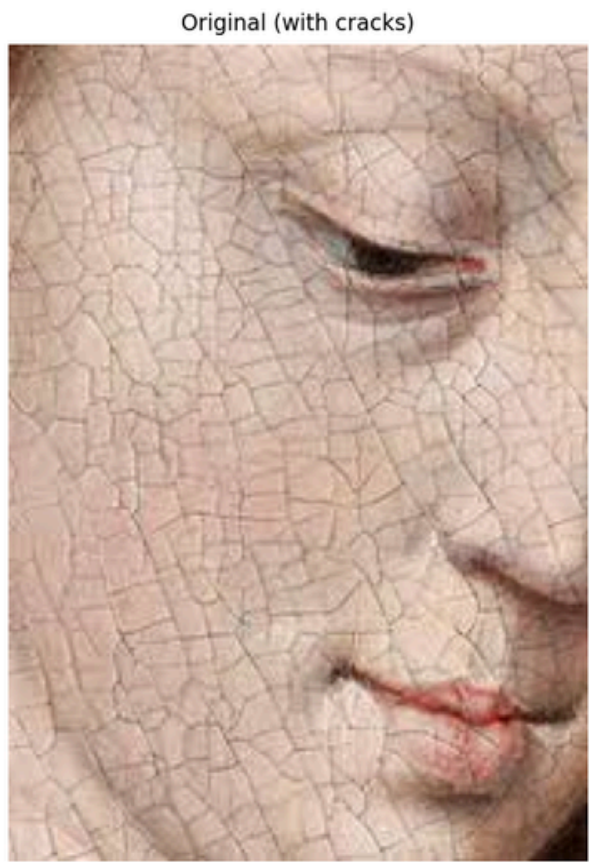
```
Crack pixel p → Find best patch Q (non-crack) → Copy texture from Q → Fill p
```


Comparison: Classical vs Modern vs Novel

Method	PSNR (dB)	SSIM	Key Features
Classical AD (2006)	30.87	0.9045	Local Diffusion
Modern AD (2026)	30.95	0.90611	Orientati on Aware
Novel EBPI	30.29	0.8979	Texture patch copy

- Modern AD consistently outperforms Classical across all 4 paintings.
- EBPI trades PSNR for perceptual texture consistency.
- All methods use same crack mask comparison is purely inpainting quality.

Classical vs Modern vs Novel EBPI | painting3
Exemplar-Based Patch Inpainting (EBPI) — Proposed Novel Method



THREE-WAY RESULTS	
=====	
Classical AD:	
PSNR=30.8705 dB	
SSIM=0.9045	
Modern Orient-AD:	
PSNR=30.9518 dB	
SSIM=0.9061	
Novel EBPI [Proposed]:	
PSNR=30.2856 dB	
SSIM=0.8979	
EBPI vs Classical AD:	
PSNR: -0.5849 dB	
SSIM: -0.0066	
MSE: +7.6720	

Fig. Threeway Comparison of methods

Thank

You